

Medial Patellofemoral and Medial Patellotibial Ligament Reconstruction Using a Docked Rectus Femoris Graft

Felipe Silveira Martins, M.D.,¹  Augusto Thadeu Cardoso Filho, M.D.,¹ 
 Sergio Marinho de Gusmão Canuto, M.D.,²  Camilo Partezani Helito, M.D., Ph.D.,³ 
 Bernardo Garcia Barroso, M.D.,⁴  Márcio Cabral Fagundes Rêgo, M.D.,⁵ 
 Marcelo Cabral Fagundes Rêgo, M.D.,⁵  and Diego Ariel de Lima, M.D., Ph.D.^{6*} 

Abstract: Recurrent patellar instability may persist after isolated medial patellofemoral ligament reconstruction, particularly in cases with instability beyond early knee flexion. Recent evidence has emphasized the role of the medial patellotibial ligament as a secondary stabilizer contributing to patellofemoral stability at higher degrees of flexion. As a result, combined reconstruction of the medial patellofemoral ligament and medial patellotibial ligament has emerged as a promising strategy to restore more physiologic patellar kinematics. This technical note describes a reproducible technique for combined medial patellofemoral ligament and medial patellotibial ligament reconstruction using a single rectus femoris tendon autograft harvested from the superficial layer of the quadriceps tendon. The technique is soft-tissue-based and physeal-sparing and avoids bone tunnels, making it particularly suitable for skeletally immature patients. The rectus femoris graft provides sufficient length, favorable biomechanical properties, low donor-site morbidity, and satisfactory cosmetic results through a minimally invasive approach.

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Patellar instability is a multifactorial condition that mainly affects young and active patients and is associated with recurrent dislocation, pain, functional impairment, and progressive patellofemoral cartilage damage.^{1,2} The medial patellofemoral ligament (MPFL) is the primary passive restraint against lateral patellar translation, providing approximately 50% to 60% of medial stability in the first 20° to 30° of knee flexion.^{2,3}

Isolated MPFL reconstruction has shown good clinical outcomes; however, residual instability or recurrent dislocation may persist, particularly in patients with patella alta, increased tuberosity-trochlear groove distance, ligamentous laxity, or instability occurring beyond early knee

flexion.^{4,5} Increasing evidence highlights the role of secondary medial stabilizers, especially the medial patellotibial ligament (MPTL), which contributes significantly to patellar stability at higher degrees of knee flexion and influences patellar tilt and rotation.^{2,6}

Biomechanical and clinical studies suggest that combined reconstruction of the MPFL and MPTL more closely restores native patellofemoral kinematics and may reduce stress on the reconstructed MPFL compared with isolated MPFL reconstruction.^{7,8} Nevertheless, most described techniques rely on hamstring or patellar-based grafts and may involve multiple tunnels, increased donor-site morbidity, or technical complexity.⁹

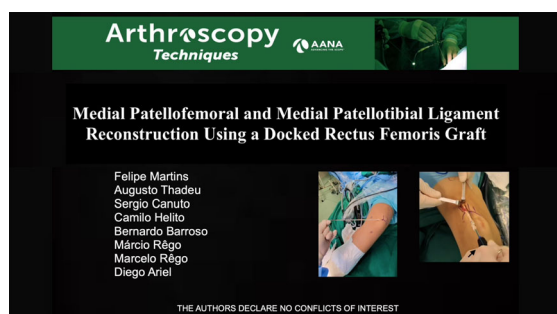
From the ¹Clinica de Ortopedia e Traumatologia (COT), Balneário Camboriú, Brasil; ²Ortopedia Hospital de Ortopedia, Maceió, Brasil; ³Instituto de Ortopedia e Traumatologia, Universidade de São Paulo, São Paulo, Brasil; ⁴Instituto de Ortopedia e Traumatologia, Vitória Apart Hospital, Serra, Brasil; ⁵Hospital Memorial São Francisco, Natal, Brasil; ⁶Universidade Federal Rural do Semi-Árido (UFERSA), Mossoró, Brasil.

*Address correspondence to Diego Ariel de Lima, M.D., Ph.D., Universidade Federal Rural do Semi-Árido (UFERSA), R. Francisco Mota, 572 - Pres. Costa e Silva, Mossoró - RN, CEP: 59625-900, Brazil.
 E-mail: arieldelima.diego@gmail.com

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VIDEO 1 The video shows the surgical technique for combined medial patellofemoral and medial patellotibial ligament reconstruction using a rectus femoris tendon graft, performed in a left knee. Video content can be viewed at <https://doi.org/10.1002/atn2.70252>.

In parallel, the rectus femoris tendon harvested from the superficial layer of the quadriceps has emerged as a versatile and reliable autograft in knee ligament surgery, offering consistent length, favorable biomechanical properties, and a safe harvest plane that preserves deeper quadriceps layers.¹⁰⁻¹²

The technique describes a reproducible method for combined MPFL and MPTL reconstruction using a single rectus femoris tendon autograft, aiming to restore patellar stability throughout knee motion while minimizing donor-site morbidity and surgical complexity.

SURGICAL TECHNIQUE

The complete technique is shown in Video 1. Pearls and pitfalls are shown in Table 1 and the advantages and disadvantages in Table 2.

Patient Position and Clinical Examination

The surgical technique is indicated for patients with recurrent patellar instability refractory to conservative treatment, particularly in skeletally immature patients. Preoperative clinical assessment typically shows lateral patellar instability, which is confirmed under anesthesia by patellar translation or dislocation during standardized instability testing. Imaging evaluation commonly includes magnetic resonance imaging to assess trochlear morphology, patellar tracking, joint effusion, and integrity of the MPFL, as well as computed tomography using a standardized protocol to evaluate patellar height and tuberosity-trochlear groove distance.

The patient is positioned supine on the operating table under general anesthesia. A pneumatic tourniquet is applied to the proximal thigh. Examination under anesthesia is performed to confirm patellar instability and to evaluate patellar tracking throughout the knee range of motion.

TABLE 1 Pearls and Pitfalls

Pearls	Pitfalls
1. Precise identification of the surgical plane between the superficial rectus femoris tendon and the deeper quadriceps layers is essential to preserve the extensor mechanism 2. Careful and progressive dissection of the superficial rectus femoris tendon before introducing the tendon stripper reduces the risk of graft amputation 3. Maintaining the distal attachment of the rectus femoris tendon to the patella facilitates graft handling, orientation, and controlled tensioning 4. Development of a subretinacular, extra-articular plane between the medial retinaculum and the joint capsule is critical for anatomic graft passage without intra-articular violation 5. Fluoroscopic confirmation of Schöttle's point is recommended to ensure accurate femoral fixation and to avoid physeal injury in skeletally immature patients 6. Graft tensioning should allow physiologic patellar mobility, with assessment of patellar tracking throughout the full range of knee motion	1. Incomplete or imprecise dissection of the rectus femoris tendon may lead to graft amputation or violation of deeper quadriceps layers 2. Excessive force applied during advancement of the tendon stripper may compromise graft integrity or cause premature proximal release 3. Incorrect identification of the femoral attachment point may result in graft malposition, overconstraint, or increased medial patellofemoral joint pressure 4. Failure to use fluoroscopy for femoral or tibial fixation increases the risk of physeal injury in skeletally immature patients 5. Overtensioning of the graft during MPFL or MPTL fixation may lead to restricted patellar mobility, medial tilt, or postoperative stiffness 6. Nonanatomic routing or fixation of the graft to the tibia may result in ineffective MPTL reconstruction or soft-tissue irritation
MPFL, medial patellofemoral ligament; MPTL, medial patellotibial ligament.	

TABLE 2 Advantages and Disadvantages

Advantages	Disadvantages
1. Use of a single rectus femoris tendon autograft provides sufficient length and versatility to reconstruct both the MPFL and MPTL through a single harvest site 2. The technique is soft-tissue-based and physal-sparing, making it particularly suitable for skeletally immature patients 3. Preservation of the deeper quadriceps layers minimizes extensor mechanism morbidity while providing a graft with reliable biomechanical properties 4. Low surgical morbidity and favorable cosmetic outcome are achieved through a minimally invasive approach using only 4 small incisions	1. Harvesting the rectus femoris tendon requires familiarity with quadriceps anatomy and careful dissection to avoid graft damage 2. There is a potential risk of donor-site hematoma or transient quadriceps weakness in the early postoperative period 3. Precise graft tensioning is technically demanding and essential to avoid patellar overconstraint or residual instability 4. Long-term clinical outcomes of combined MPFL and MPTL reconstruction using a rectus femoris graft remain limited in the literature

MPFL, medial patellofemoral ligament; MPTL, medial patellotibial ligament.

Arthroscopic Evaluation

A standard diagnostic arthroscopy is performed using anterolateral and anteromedial portals. The patellofemoral joint is inspected to confirm lateral patellar maltracking and instability. Articular cartilage surfaces of the patella and trochlea are evaluated, and associated intra-articular pathology is ruled out or addressed as indicated. Arthroscopic findings are correlated with preoperative imaging to confirm the indication for combined MPFL and MPTL reconstruction.

Necessary Materials for the Procedure

The following materials are used in this procedure: standard knee arthroscopy instrumentation, including a 30° arthroscope, probe, and shaver system (Arthrex, Naples, FL); an open-loop tendon stripper for soft-tissue graft harvesting (ConMed Linvatec, Largo, FL); an intraoperative fluoroscopy unit; absorbable sutures for temporary fixation and soft-tissue closure (Vicryl; Ethicon, Somerville, NJ) and high-strength nonabsorbable sutures for graft fixation (FiberWire; Arthrex); 4-mm bioabsorbable suture anchors for femoral, patellar, and tibial fixation (BioComposite SutureTak; Arthrex); and standard orthopaedic surgical instruments for soft-tissue dissection and fixation, including scalpels, forceps, Backhaus clamps, needle holders, retractors, Metzenbaum scissors, blunt dissectors, and electrocautery devices.

Graft Harvest and Preparation of the Graft

Three reference lines are marked on the superior pole of the patella to delineate the lateral, intermediate, and medial portions of the quadriceps tendon. A longitudinal

incision is made between the lateral and intermediate lines. Superficial dissection is carefully performed to expose the quadriceps tendon (Figure 1).

A strip of the rectus femoris tendon, approximately 7 mm in width, is identified and isolated from the superficial layer of the quadriceps tendon while preserving the intermediate and deep layers. The rectus femoris tendon is maintained attached distally to the patella. Using Metzenbaum scissors, dissection is extended proximally toward the musculotendinous junction, which is subsequently released.

The graft is tensioned using a nonabsorbable traction suture, and blunt dissection is facilitated with a clamp. An open-loop tendon stripper is advanced proximally to release the rectus femoris tendon up to its proximal insertion. The free end of the graft is prepared with absorbable sutures. The harvested graft typically measures approximately 24 to 26 cm in length, allowing reconstruction of both medial stabilizers (Figures 2 and 3).

MPFL Reconstruction

A medial parapatellar incision is performed, creating an extra-articular plane, and the graft is passed through this plane. Initial fixation is performed at the proximal patellar attachment site of the MPFL using absorbable sutures (Figure 4).

A separate medial incision is made over the adductor tubercle, which serves as an anatomical reference. A guidewire is positioned at the femoral attachment point of the MPFL (Schöttle's point¹³), and correct placement is confirmed under fluoroscopic guidance to avoid physal violation. A 4-mm bioabsorbable suture anchor is inserted at the femoral attachment site (Figure 5).

The graft is passed between the medial incisions and sutured with nonabsorbable threads. Another anchor is

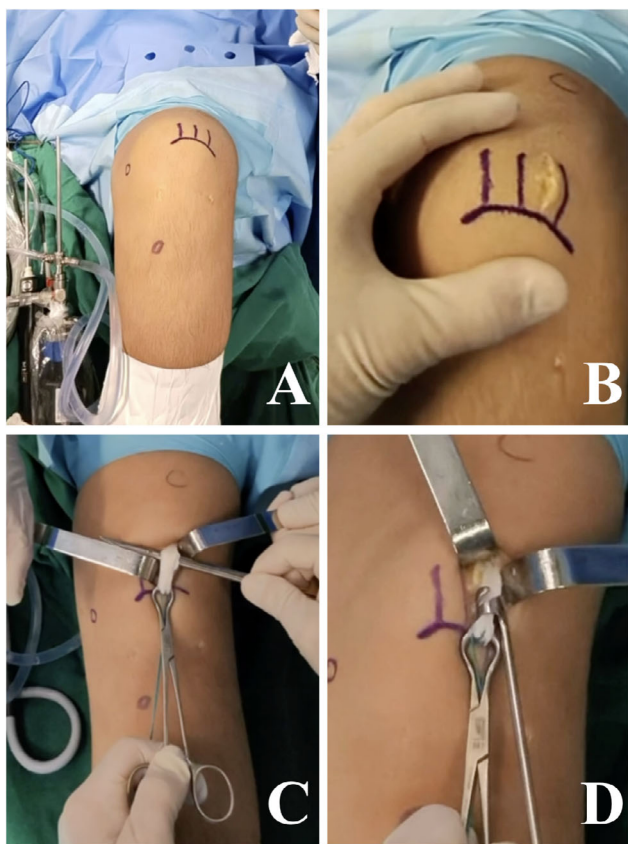


FIGURE 1 Graft harvest. The left knee is shown. (A) Three reference lines are marked on the superior pole of the patella to delineate the lateral, intermediate, and medial portions of the quadriceps tendon. (B) A 4-cm longitudinal skin incision is made between the lateral and intermediate lines. (C) Superficial dissection is carefully performed to expose the quadriceps tendon, and a strip of the rectus femoris tendon, approximately 7 mm in width, is identified and isolated from the superficial layer while preserving the intermediate and deep layers. (D) An open-loop tendon stripper is prepared for graft release.

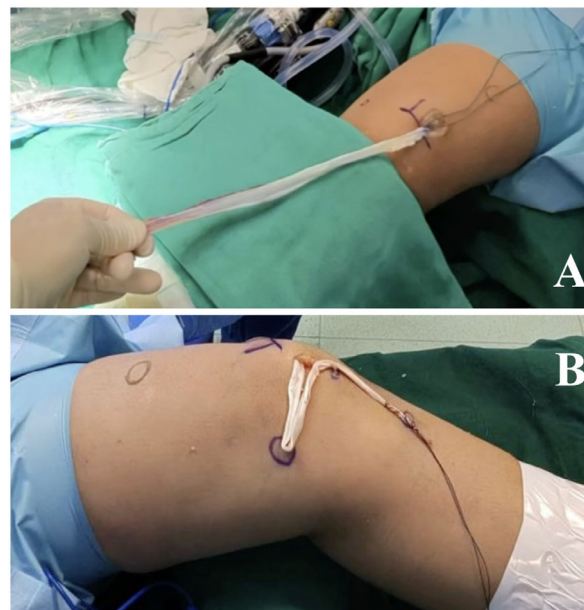


FIGURE 3 Graft harvest. The left knee is shown. (A) The rectus femoris graft remains attached to the superior pole of the patella after proximal release using an open-loop tendon stripper. (B) Final configuration of the rectus femoris graft prepared to reconstruct the medial patellofemoral and medial patellotibial ligaments.

placed at the distal patellar attachment of the MPFL in the femur, with careful attention to graft tensioning to avoid overconstraint. So, the graft is passed back to the anterior incision and secured, completing the MPFL reconstruction.

The patellar attachment of the MPFL was reconstructed using 2 fixation points along the medial border of the patella, corresponding to the native anatomical footprint. The proximal attachment was positioned at the

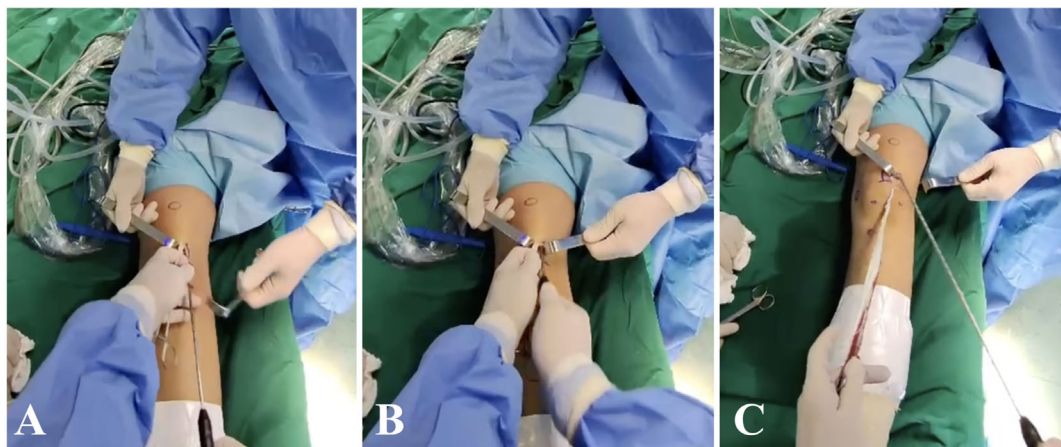


FIGURE 2 Graft harvest. The left knee is shown. (A-C) An open-loop tendon stripper is advanced proximally to release the rectus femoris tendon up to its proximal insertion.

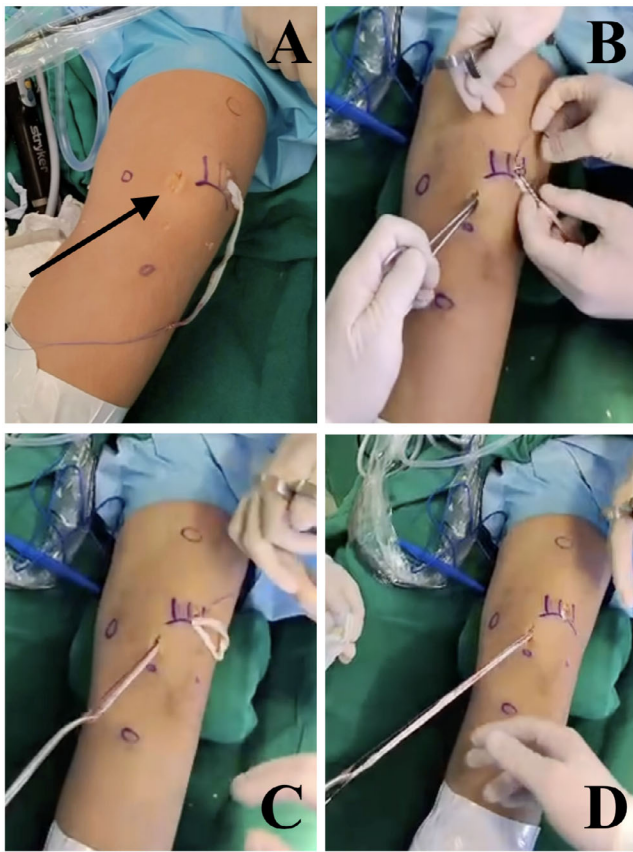


FIGURE 4 Medial patellofemoral ligament reconstruction. The left knee is shown. (A) A medial parapatellar retinacular incision is performed (black arrow). (B) Development of an extra-articular, subretinacular plane between the medial parapatellar access and the graft harvest incision. (C,D) The graft is passed through this extra-articular plane.

superomedial aspect of the patella, near the junction of the proximal third, whereas the distal attachment was placed at the midmedial patellar region. This configuration allows for a broader and more anatomic distribution of forces across the patellar insertion. The graft limb dedicated to MPFL reconstruction was passed through the second anatomical layer, between the medial retinaculum

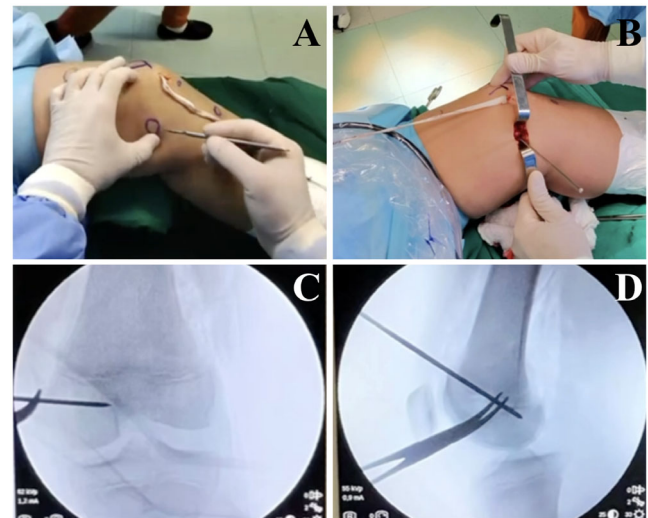


FIGURE 5 Medial patellofemoral ligament reconstruction. The left knee is shown. (A) A separate medial incision is made over the adductor tubercle. (B) A guidewire is positioned at the femoral attachment point of the medial patellofemoral ligament (Schöttle's point¹³). (C,D) Correct guidewire placement is confirmed under fluoroscopic guidance to avoid physal violation.

(layer II) and the joint capsule (layer III), reproducing the native course of the MPFL. Care was taken to maintain an extra-articular, subretinacular plane, avoiding intra-articular penetration and ensuring smooth graft excursion (Figures 6 and 7).

MPTL Reconstruction

A medial incision is performed over the proximal medial tibia. Under fluoroscopic guidance, a physal-safe metaphyseal fixation point distal to the proximal tibial physis is identified to avoid iatrogenic growth disturbance.^{14,15} The graft is passed subcutaneously from the patellar region to the tibial incision and fixed to the tibia using a 4-mm bioabsorbable suture anchor and high-strength nonabsorbable



FIGURE 6 Medial patellofemoral ligament reconstruction. The left knee is shown. (A) A 4-mm bioabsorbable suture anchor is inserted at the femoral attachment site. (B,C) The graft is passed between the medial incisions and secured using nonabsorbable sutures.

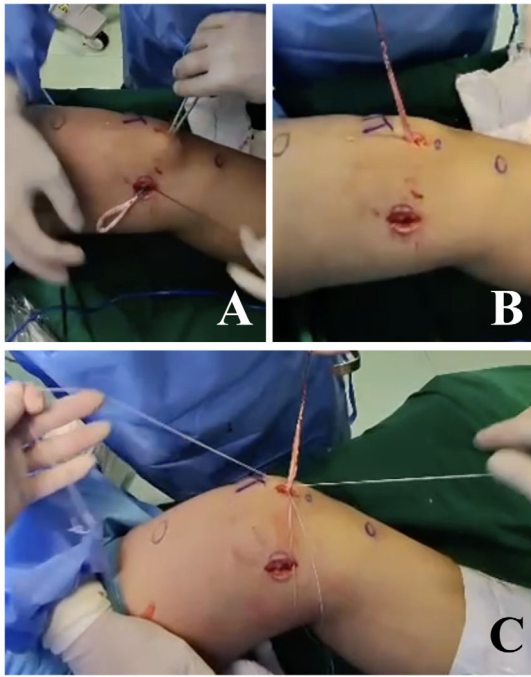


FIGURE 7 Medial patellofemoral ligament reconstruction. The left knee is shown. (A-C) The graft is passed back to the anterior incision and secured, completing the medial patellofemoral ligament reconstruction.

sutures. The graft limb was directed distally through a subcutaneous extra-articular plane from the patellar region toward the proximal medial tibia. This trajectory reproduces the native orientation of the MPLT and avoids violation

of deeper anatomical layers. Graft tension is adjusted to maintain physiologic patellar mobility throughout the full range of knee motion (Figure 8).

Final Assessment

After graft fixation, patellar tracking is assessed through the full range of knee motion. The patella should show stable, centralized tracking without excessive medial constraint or lateral instability (Figure 9). All incisions are closed in a layered and cosmetic fashion. No routine immobilization is required beyond standard postoperative protection protocols.

Postoperative Protocol

Postoperatively, partial weight-bearing with crutches is allowed as tolerated, using a hinged knee brace locked in extension during ambulation for the first 2 weeks. Early knee range of motion is encouraged under supervision, initially limited to 0° to 60°, with cryotherapy and limb elevation used to control pain and swelling. Quadriceps activation exercises are initiated within the first postoperative days.

Between weeks 3 and 6, knee flexion is gradually progressed to 90°, and weight-bearing is advanced to full as tolerated. The brace is discontinued once adequate quadriceps control is achieved. Closed kinetic chain

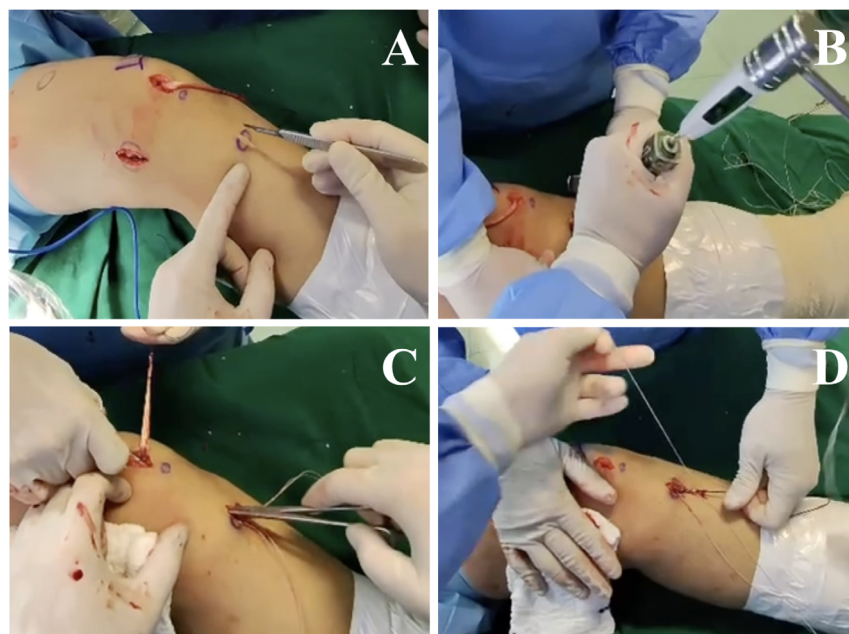


FIGURE 8 Medial patellotibial ligament reconstruction. The left knee is shown. (A) A medial incision is performed over the proximal medial tibia. (B) A bioabsorbable suture anchor is placed at a physal-safe metaphyseal fixation point. (C,D) The graft is passed subcutaneously from the patellar region to the tibial incision and fixed to the tibia.



FIGURE 9 Final assessment. The left knee is shown. Low surgical morbidity and a favorable cosmetic outcome are achieved through a minimally invasive approach using only 4 small incisions.

strengthening, proprioceptive training, and neuromuscular control exercises are progressively introduced, whereas running, jumping, and pivoting activities are avoided.

After 3 months, functional strengthening and low-impact aerobic activities are advanced. Return to pivoting sports is generally allowed after 6 months, based on clinical examination, functional recovery, and absence of patellar instability.

DISCUSSION

The MPFL is the primary restraint to lateral patellar translation in early knee flexion, and its reconstruction is the cornerstone of surgical treatment for recurrent patellar instability.^{2,3} Nevertheless, persistent instability and residual maltracking have been reported after isolated MPFL reconstruction, particularly in patients with instability beyond 30° of flexion or in the presence of anatomical risk factors.^{4,5}

Recent biomechanical and clinical evidence has emphasized the role of the MPTL as a relevant secondary stabilizer, especially at higher degrees of knee flexion.^{2,6} Combined MPFL and MPTL reconstruction has been shown to improve control of patellar translation and rotation while reducing stress on the reconstructed MPFL when compared with isolated MPFL reconstruction.^{7,8,16}

Although contemporary pediatric techniques often favor epiphyseal tibial fixation for MPTL reconstruction to better preserve graft isometry, physeal-safe metaphyseal

fixation has been historically used in pediatric patellar stabilization procedures derived from tenodesis-style techniques. These approaches aim to avoid direct physeal violation while providing a distal checkrein effect on the patella. However, metaphyseal fixation may sacrifice strict isometry and therefore requires careful intraoperative tensioning to avoid overconstraint, anterior knee pain, or limitation of flexion.^{2,14,15}

Most combined reconstruction techniques described in the literature rely on hamstring or patellar-based grafts and frequently require bone tunnels or extensive fixation, which may increase technical complexity and donor-site morbidity, particularly in skeletally immature patients.^{9,17} For this population, physeal-sparing, soft-tissue-based techniques with anchor fixation have been recommended to minimize the risk of growth disturbance.^{14,18}

The rectus femoris tendon harvested from the superficial layer of the quadriceps has emerged as a versatile autograft option in knee ligament reconstruction, providing consistent length, reliable harvest, and preservation of deeper quadriceps layers.¹⁰⁻¹² Biomechanical studies have shown that the rectus femoris graft presents structural properties comparable to the bone–patellar tendon–bone graft, with similar stiffness and ultimate load to failure.¹⁹⁻²² In addition, clinical studies have shown comparable early quadriceps strength recovery between rectus femoris and hamstring autografts, supporting its use without significant extensor mechanism compromise.²³

The technique combines the biomechanical reliability of the rectus femoris graft with an anatomic reconstruction of both the MPFL and MPTL using a single graft and limited surgical exposure. By avoiding bone tunnels, respecting physeal anatomy, and minimizing donor-site morbidity, this approach aims to restore patellar stability throughout the full range of motion while remaining reproducible and applicable in skeletally immature patients.

DISCLOSURES

The authors (F.S.M., A.T.C.F., S.M.G.C., C.P.H., B.G.B., M.C.F.R., M.C.F.R., D.A.L.) declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

ORCID

Felipe Silveira Martins  <https://orcid.org/0009-0005-8195-0036>

Augusto Thadeu Cardoso Filho  <https://orcid.org/0009-0001-5521-081X>

Sergio Marinho de Gusmão Canuto  <https://orcid.org/0000-0003-1339-4785>

Camilo Partezani Helito  <https://orcid.org/0000-0003-1139-2524>

Bernardo Garcia Barroso  <https://orcid.org/0000-0002-8447-7660>

Márcio Cabral Fagundes Rêgo  <https://orcid.org/0009-0001-2755-9969>

Marcelo Cabral Fagundes Rêgo  <https://orcid.org/0009-0009-8499-8659>

Diego Ariel de Lima  <https://orcid.org/0000-0002-5702-1529>

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